



Mayank Jha
Computer Science & Engineering
Indian Institute of Technology Bombay

25M0762
M.Tech.
Gender: Male
DOB: 23/09/2002

Examination	University	Institute	Year	CPI / % Credits
Post Graduation	IIT Bombay	IIT Bombay	2027	null
Graduation	IIIT Delhi	IIIT Delhi	2024	7.11
Graduation Specialization: Computer Science and Biosciences				

M.TECH PROJECT AND SEMINAR

Game Theoretic Approaches for Security Planning

July 2026 - Present

Prof. Swaprava Nath, Dept. of CSE, M.Tech Project

IIT Bombay

- Synthesized 3 game-theoretic frameworks from 28 sources on SAM-SEAD planning, **translating technical models into actionable insights**.
- Modeled allocation for 22 aircraft across 10 barriers**, benchmarking 4 baselines and implementing patrol schedules to maximize security.

PROJECTS

Real-Time Order Matching Engine

Jan 2026 – May 2026

Prof. G Sivakumar, Dept. of CSE, Course Project

IIT Bombay

- Built an automated trading system** that matched MARKET and LIMIT orders using price-time priority, handling partial fills and processing **4.72M orders/sec** with 42 ns p50 latency on a **500K-order book**
- Connected the trading platform end-to-end** with a Rust/Tokio matching service, REST-to-gRPC communication, a React dashboard, and secure user authentication, allowing orders to flow from the web interface through the matching engine

Full-Stack Search Engine for a Large Document Corpus

July 2026 - Present

Prof. Soumen Chakrabarti, Dept. of CSE, Course Project

IIT Bombay

- Designed a search engine** for 19,500+ documents, evaluating 35 ranking and retrieval configurations to achieve a 20.1% relevance gain.
- Spearheaded a 3-person team** to build automated checks that exposed 58× selection bias in relevance judgments pre-deployment.

Database Optimization Decision-Support Tool

July 2026 - Present

Prof. S. Sudarshan, Dept. of CSE, Course Project

IIT Bombay

- Engineered a database decision tool using execution benchmarking, **achieving up to 99.2% faster query execution**.
- Implemented a greedy optimization algorithm, **eliminating up to 33% of redundant index candidates** across multi-query workloads.

TECHNICAL SKILLS

Languages & Algorithms: Python, C++, SQL, JavaScript, Java, Data Structures & Algorithms, Dynamic Programming, Optimization

Data & Analytics: Pandas, NumPy, Data Visualization, Statistical Analysis, Data Cleaning, Exploratory Data Analysis

Machine Learning: scikit-learn, PyTorch, Natural Language Processing, Deep Learning, Embeddings, Model Evaluation, TF-IDF

Databases & Engineering: PostgreSQL, MySQL, Redis, Query Optimization, Indexing, REST APIs, Flask, Node.js, Git, Linux, Docker

POSITIONS OF RESPONSIBILITY

Teaching Assistant, CS101: Computer Programming and Utilization, IIT Bombay

Aug 2025 – Nov 2025

Personally Guided 6 students through C++ fundamentals via doubt-clearing sessions resulting in improved course performance.

Teaching Assistant, CS218: Design and Analysis of Algorithms, IIT Bombay

Jan 2026 – May 2026

Prepared 4 quizzes, designed rubrics, supervised examinations, and graded 100+ evaluation papers to support course assessment.

Teaching Assistant, CS6001: Game Theory & Algorithmic Mechanism Design, IIT Bombay

July 2026 – Present

RELEVANT COURSEWORK

IIT Bombay: Software Engineering, Indexing and Retrieving Text and Graphs*(ongoing course), Foundations of Machine Learning

IIIT Delhi: Data Structures and Algorithms, Game Theory, Database Management System, Operating Systems, Artificial Intelligence,

Money & Banking, Microeconomics, Macroeconomics

EXTRACURRICULARS

Community Work with E-Vidyaloka, Sports & Fitness: Recreational Sports and Regular Strength Training

HOBBIES AND INTERESTS

Playing Chess, Reading, Learning Philosophy & Psychology, Exploring Scriptures, Spirituality & Self-Reflection, Exploring Game Theory, International Relations, Economics & History, Building Tools to Solve Everyday Problems, Cinephile, Music Enthusiast